

# **Course Syllabus**

**DATA 422: Advanced Data Science Methods and Ethics**

# DATA 422: Advanced Data Science Methods and Ethics

Cal Poly Humboldt

**Term note:** Meeting times, rooms, and office hours below are from the **Fall 2025** syllabus. Update them for the current term before publishing.

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## Instructor

**Instructor:** Dr. Rosanna Overholser

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Please use your Cal Poly Humboldt email and check Canvas regularly for course communications.

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## Meeting times and student hours (Fall 2025)

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|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Lecture       | MWF 12:00–12:50 PM in BSS 308                                                                                                                 |
| Lab           | Tuesday 11:00 AM–12:50 PM in BSS 313                                                                                                          |
| Student hours | Monday 2:00–3:50 PM and Wednesday 1:00–2:50 PM in BSS 334 (may move to BSS 312 if needed). Happy to meet by appointment—email to find a time. |

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## Catalog description

This course provides a treatment of advanced topics in Data Science. Topics include: (i) Privacy and Ethics, (ii) Multivariate Statistics, (iii) Big Data and Cloud Computing, and (iv) Deep Learning. The course will draw from numerous case studies and applications, with a practical rather than theoretical emphasis.

A minimum grade of **C-** is required to count toward the Data Science major.

**Prerequisites:** DATA 322.

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## Required materials

- **Textbook:** *The Hundred-Page Language Model Book* (free online), plus readings provided in Canvas.
  - **Technology:** A computer for work outside lab (personal, Data Science program, or library checkout). If you need a laptop, consider Cal Poly Humboldt's Reusable Office Supply Exchange (R.O.S.E.) program.
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## Course load

DATA 422 has 3 units of lecture and 1 unit of activity. Expect about **5 hours in class** and at least **7 hours out of class** each week.

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## Assessment and grading

| Component | Weight |
|-----------|--------|
| Quizzes   | 40%    |
| Project 1 | 20%    |
| Project 2 | 20%    |
| Project 3 | 20%    |

## How assessments work

- There are **no midterms** and **no written final exam**.
- **Project 3** includes a presentation scheduled in the **final-exam time slot**.
- **Stepping stones** scaffold the projects (lab demos, checkpoints, weekly question sets). They are **not** a separate grade weight; they support the project grades.
- **Weekly quizzes** draw on that week's **lab, lecture**, and assigned **reading or videos**.

Letter grades (nearest percent): 93–100 (A); 90–92 (A-); 87–89 (B+); 83–86 (B); 80–82 (B-); 77–79 (C+); 73–76 (C); 70–72 (C-); 60–69 (D); below 60 (F).

See [Assessments](#) for links as materials are posted.

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## Course structure

Meetings include discussions and small-group activities based on readings distributed beforehand. Projects practice applying course content. Typical project deliverables include shared code of ethics with reflection, a presentation (Project 3 / final slot), and a written report with code appendix—exact prompts will be posted under [projects](#).

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## Missed or late work

Policies are discussed and agreed upon in the first two weeks of class, then posted on Canvas.

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## Academic integrity

Students are responsible for knowing and abiding by Cal Poly Humboldt's academic honesty policy. Students caught cheating will receive zero points and be reported to Student Judicial Affairs. Working in small groups on projects and when preparing for quizzes is encouraged, but work submitted for grading must be your own.

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## AI policy

You may use AI (for example ChatGPT or Claude) to help you learn the materials, perform data analyses, or work on projects. For graded work completed outside class:

1. If you used AI, acknowledge it briefly in the document (name the service; detailed prompts are not required).
2. You are responsible for the correctness and appropriateness of everything you turn in (AI-assisted or not).

The instructor may use AI to improve course materials (for example checking whether an assignment is college-level or clarifying quiz wording). AI will **not** be used to grade your work, and AI detection tools will not be used to check your work.

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## Campus policies and resources

[Syllabus addendum: campus resources and policies](#)

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## Learning outcomes

### Program / student learning outcomes (summary)

Students will demonstrate computational skills to extract and visualize data; statistical knowledge to build models and ensure valid analysis; contemporary data-oriented skills with ethical considerations; and effective communication for diverse audiences. Measurable behaviors include extracting and summarizing data from large public databases, designing models that guide valid analysis, identifying frameworks and biases relevant to a dataset, and preparing written descriptions of methodology and implications.

## Course learning outcomes

Upon successfully completing this course, you will be able to:

- Implement practical and scalable ways for multiple parties to share data, models, and model outcomes while preserving the privacy of each party's data.
- Explain how machine learning algorithms interpret data and the potential for biases in these algorithms.
- Apply multivariate statistical approaches such as principal component analysis and tests of significance to data sets.
- Work with a big data set using a cloud computing system.
- Build a deep learning model and articulate what distinguishes deep learning from other machine learning algorithms.